SMARTBRIDGE
Let's Bridge the Gap

Project Initialization and Planning Phase

Date	21/10/2026
Team ID	Suhani Porwal
Project Title	**Electric Vehicle Market Performance & Insights**
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Analyse EV market performance and present clear insights on sales battery technologie driving range, and customer satisfaction.
Scope	Explore available EV market data, clean and prepare it, calculate key indicators, and build visual reports or an interactive dashboard. Conclusions are limited to the fields and period available in the dataset.
Problem Statement	
Description	EV market information is spread across multiple metrics and vehicle segments, making it difficult to compare manufacturers, technologies, regions, and customer feedback in one place.
Impact	A consolidated view can reduce manual comparison effort, make trends easier to identify, and support market planning, product positioning, and sustainable-mobility research.
Proposed Solution	
Approach	Import the dataset, validate and clean records, standardize categories, analyze KPIs, an present findings with charts, KPI cards, filters, and comparisons.
Key Features	Annual EV sales analysis; manufacturer and market-share comparison; battery technology distribution; driving-range and battery-capacity comparison; manufacturing-country analysis; customer-rating insights.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computer or laptop for data processing and dashboard development.	Standard laptop/PC; CPU and RAM depend on dataset size.
Memory	RAM for loading, cleaning, and analyzing the dataset.	**8 GB RAM recommended**; **16 GB helpful** for larger datasets.
Storage	— Space for raw data, cleaned data, project files, reports, and backups	At least **5–10 GB free space**, depending on dataset size.
Software		
Frameworks	Tool to clean data, calculate metrics, and create charts or dashboard.	Power BI, Tableau, Excel, or Python libraries—use the tool actually selected.
Libraries	Optional libraries for data cleaning and analysis if using Python.	Pandas, NumPy, Matplotlib (only if Python is used).
Development Environment	Application used to edit, analyze, and manage project files.	Power BI Desktop / Excel / Jupyter Notebook / VS Code, as applicable.
Data		
Data	Source and fields for sales, manufacturer, vehicle type, battery specifications, range, country, and customer ratings.	CSV/Excel or other available dataset; record source, time period, row count, and data limitations.